Optical Transport Failure Resolution SOP for Cisco PE KOR-PE-01

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_3101

Description

This Standard Operating Procedure (SOP) outlines the steps for resolving Optical Transport Failures on Cisco PE equipment, specifically Node ID KOR-PE-01 located in the Koramangala Datacenter. The failure domain in question pertains to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, as indicated by Alarm Code ALARM_3101 with a severity of Minor. The underlying protocol or hardware mechanics responsible for this failure domain involve the Optical Transport Network (OTN) protocol, which is used for transporting client traffic over optical fiber. The OTN protocol operates at the physical layer, employing SFP/XFP transceivers to transmit and receive optical signals. In the event of SFP/XFP degradation or fiber cuts, the OTN protocol may experience errors, leading to the generation of Alarm Code ALARM_3101. Therefore, this SOP aims to troubleshoot and resolve the root cause of the failure, ensuring minimal disruption to the network.

Pre-requisite Checks

- Dispatch remote hands to the Koramangala Datacenter for physical access to the equipment.
- Verify the status of redundant links to ensure network availability.
- Confirm that the Cisco PE equipment is running the latest firmware version.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP Transceivers	High
Fiber Cuts or Damages	Medium
Rx/Tx Power Loss due to Optics or Fiber Issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface transceiver, Expected Output: Transceiver information, including temperature, voltage, and laser bias current., Description: This command displays the transceiver's status, allowing the technician to identify any issues with the SFP/XFP.

Command: show inventory, Expected Output: Inventory information, including part numbers and descriptions of installed hardware., Description: This command provides a list of installed hardware, including the SFP/XFP transceivers, enabling the technician to verify the presence of the suspected faulty component.

Command: show run | include optics, Expected Output: Optics configuration, including settings for SFP/XFP transceivers.,
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Description: This command displays the optics configuration, allowing the technician to verify the settings for the SFP/XFP transceivers and identify any potential issues.
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Resolution Steps

Step 1: Step: Replace Degraded SFP/XFP Transceivers, Command: replace slot 1/1 transceiver, Description: Replace the degraded SFP/XFP transceiver with a new one, ensuring that the new transceiver is properly seated and secured.

Step 2: Step: Verify and Repair Fiber Cuts or Damages, Command: verify fiber integrity, Description: Verify the integrity of the fiber cable and repair any cuts or damages found. Use the show interface transceiver command to verify that the fiber integrity has been restored.

Step 3: Step: Adjust Rx/Tx Power Levels, Command: configure optics, Description: Adjust the Rx/Tx power levels to compensate for any issues with the optics or fiber. Use the show run | include optics command to verify that the adjustments have been applied correctly.

Post-Check Verification

```
Command: show interface transceiver, Expected Output: Transceiver information, including temperature, voltage, and laser bias current., Description: Verify that the transceiver's status has improved after the resolution steps have been applied.

Command: show inventory, Expected Output: Inventory information, including part numbers and descriptions of installed hardware., Description: Verify that the new SFP/XFP transceiver has been properly installed and recognized by the system.

Command: show run | include optics, Expected Output: Optics configuration, including settings for SFP/XFP transceivers., Description: Verify that the optics configuration has been updated correctly after the resolution steps have been applied.
```

Roll-back Procedures

- Step: Restore Original SFP/XFP Transceivers, Command: replace slot 1/1 transceiver original, Description: Restore the original SFP/XFP transceiver that was replaced during the resolution steps, ensuring that the system returns to its original state.

- Step: Revert Optics Configuration, Command: configure optics revert, Description: Revert the optics configuration to its original state, ensuring that the system returns to its original settings.